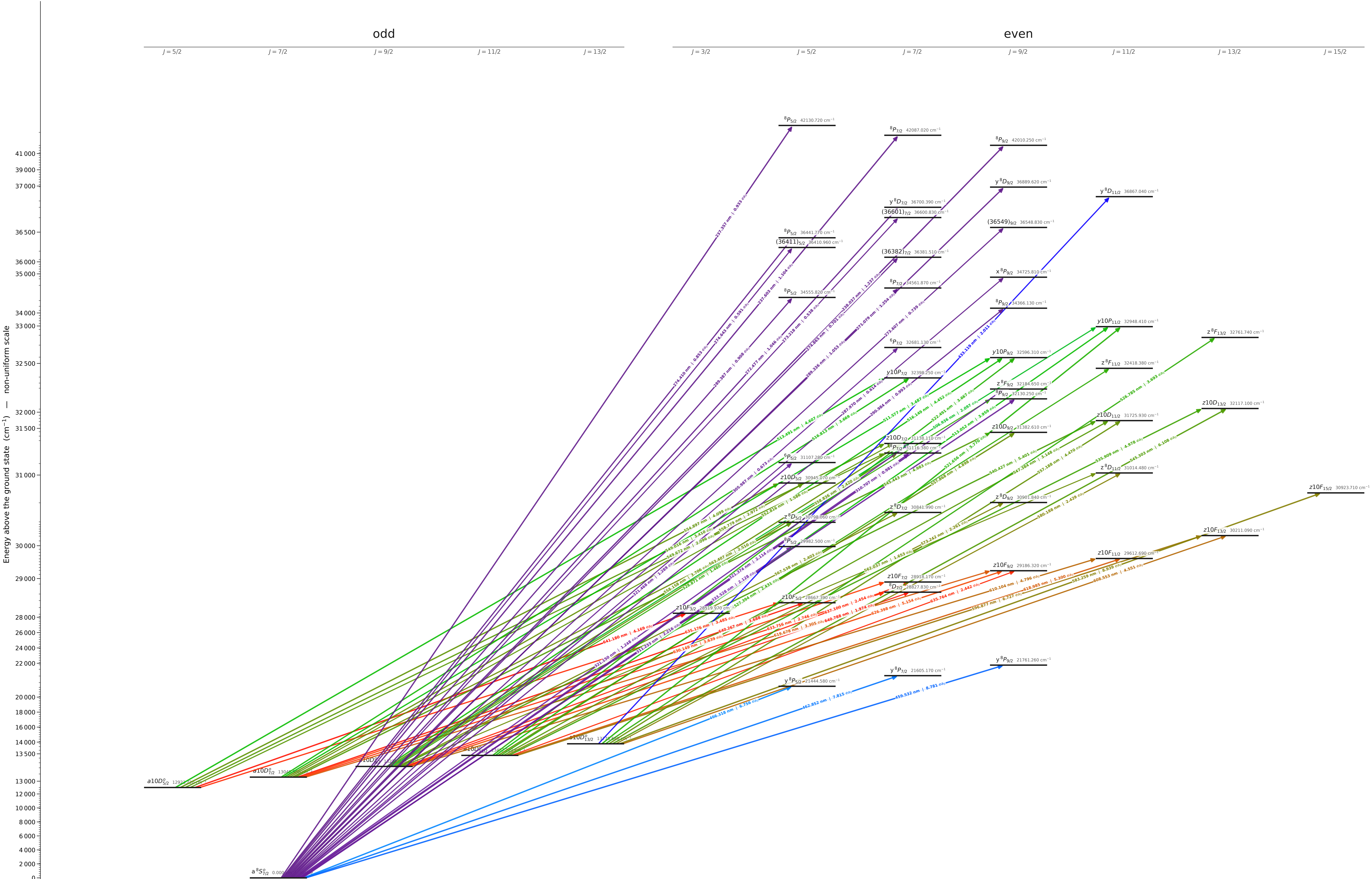
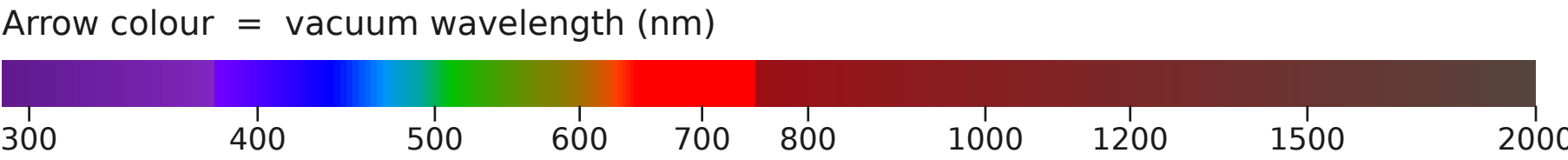


Europium

Eu (Eu I). Strongest classified transitions below 2 μm (NIST): 52 levels, 70 lines.

Eu⁺ ionisation limit 45734.74 cm⁻¹ = 5.67039 eV (218.652 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[U||er||J']$ in eao
Matrix elements are measured or NIST values; * = high-accuracy theory, = = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U er J']$ (eao)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁸ S _{7/2} - ⁸ P _{9/2}	311.2332	311.1429	963.2407	2.216	3.300e+07	-	-	-
a ⁸ S _{7/2} - ⁸ P _{7/2}	321.3741	321.2813	932.8456	2.114	3.410e+07	-	-	-
a ⁸ S _{7/2} - ⁸ P _{5/2}	333.5279	333.4320	898.8527	2.128	4.120e+07	-	-	-
a ⁸ S _{7/2} - y ⁸ P _{9/2}	459.5322	459.4035	652.3862	8.781	1.610e+08	-	-	-
a ⁸ S _{7/2} - y ⁸ P _{7/2}	462.8522	462.7226	647.7067	7.815	1.560e+08	-	-	-
a ⁸ S _{7/2} - y ⁸ P _{5/2}	466.3183	466.1878	642.8923	6.756	1.520e+08	-	-	-
a10D _{5/2} - y10P _{7/2}	513.4912	513.3482	583.8317	4.607	3.970e+07	-	-	-
a10D _{5/2} - y10P _{9/2}	516.1485	516.0047	580.8260	4.452	2.920e+07	-	-	-
a10D _{13/2} - y10P _{11/2}	521.6558	521.5105	574.6940	5.770	3.960e+07	-	-	-
a10D _{9/2} - z10D _{11/2}	540.4269	540.2767	554.7327	5.401	3.120e+07	-	-	-
a10D _{13/2} - z10D _{13/2}	545.3032	545.1517	549.7720	6.108	3.330e+07	-	-	-
a10D _{7/2} - z10D _{9/2}	545.4433	545.2918	549.6308	4.983	3.100e+07	-	-	-
a10D _{5/2} - z10D _{5/2}	554.8974	554.7433	540.2665	4.099	3.320e+07	-	-	-
a10D _{3/2} - z10F _{15/2}	583.2594	583.0977	513.9951	8.939	5.100e+07	-	-	-
a10D _{11/2} - z10F _{13/2}	596.8767	596.7114	502.2687	6.727	3.080e+07	-	-	-
a10D _{5/2} - z10F _{9/2}	641.1798	641.0026	467.5638	4.169	3.340e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).